

Antifouling on Gecko's Feet Inspired Fibrillar Surfaces: Evolving from Land to Marine and from Liquid Repellency to Algae Resistance

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In nature, biological creatures (including plants and animals) have self-cleaning capability despite the vagaries of the environment, i.e., from sky to land, and then to marine and vagaries of foulants (non-living and living). Gecko's feet have the general self-cleaning property both in air and underwater so as to keep their feet all clean for traveling through changing their adhesion. The present work reports Gecko's fibrillar structures to demonstrate general antifouling property in air through hydrophobicity and underwater after modification with hydrophilic polymer brushes. Fibrillar polypropylene (PP) nanoarrays are fabricated by hot embossing, exhibiting superhydrophobic antifouling in air. By grafting hydrophilic polyelectrolyte brushes (PSPMA) via surface-initiated atom transfer radical polymerization, they show superoleophobic antifouling of oil droplet and algae adhesion underwater. The effect of the structure of PP nanofiber arrays on the wettability and adhesion behavior is evaluated in detail. The results provide an important scientific principle for fabricating self-cleaning low-fouling materials with micro/nanostructure, with the hydrophobic ones being more applicable in air and the hydrophilic ones well suited underwater.

Rose petals is superhydrophobic but with high adhesion for water droplet.^[9,10] The gecko's feet act as one of the most famous natural superhydrophobic surfaces with a high water adhesion.^[11–13]

The fundamental strategy of designing superhydrophobic surfaces in air is constructing surface with appropriate micro/nanoroughness, combining modification with low surface energy substances.^[14,15] However, in fact, the superhydrophobic surfaces existing on the land are not well suitable for the underwater antifouling. For example, the white feathers of sea-birds show well self-cleaning in air but frequently contaminated by the leaked oil in ocean when pecking. By contrast, fish could keep their body clean underwater without being contaminated by oil pollution, exhibiting self-cleaning and antifouling. Study showed that the fish skin consisting of calcium phosphate, protein,^[16] and a thin layer of mucus

grants surface superoleophobicity in water with low friction coefficient.^[17] The self-healing of the fish underwater can be attributed to synergistic effect of the chemical composition and surface microtopography.^[18] Even though many excellent works on designing artificial self-cleaning superhydrophobic and superoleophobic structured surfaces have been reported, systematically investigation is still lacking of synergy considering varied surface's topography and physicochemical properties. More important, the self-cleaning properties of these functional surfaces against living marine organisms are still a challenge due to complicated/diversified ocean environment. We restrict our idea here and try to explore the underlying relations between the organism's fouling and the interfacial wetting, topography, physicochemical properties, which would provide inspiration for developing new marine antifouling materials. Polypropylene (PP), as a low surface energy but high-elastic-modulus polymers, has been widely used to fabricate gecko-like fibers array,^[19] which could bear high shear forces and enable to perform strong frictional adhesion on solid surface.^[20] So this bioinspired polymer fibers array was chosen as favorable surface for experiment investigation. Here, four kinds of PP nanofibers arrays with different structure oriented have been prepared. These structured array surfaces present a gradient wettability evolution to water drop in air (high adhesion to self-cleaning), due to varied fibers interval and special

1. Introduction

In the past decades, designing of functional materials with special interface wettability properties by biomimicry^[1] has attracted great attention of material scientists.^[2–5] Superhydrophobic surfaces can be widely found in plant and animal kingdoms, in order to adapt themselves to their environment, plants or insects have developed special wetting properties such as self-cleaning, anisotropic wettability, high-adhesion to water, or antifouling. In sky, wings of various insects and some birds (e.g., legs of water striders,^[6] butterfly.^[7] show superhydrophobic properties. On the ground, lotus leaf surfaces exhibit the unusual superhydrophobicity and self-cleaning property.^[8]

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fibers top shape. After grafting hydrophilic polyelectrolyte brushes (PSPMA) via the surface-initiated atom transfer radical polymerization (SI-ATRP) using bioinspired dopamine initiator,^[21] these functionalized arrays showed stable superoleophobic in water under external force compression. Interestingly, the PNIPAAm-modified composite array exhibited reversible switching of oil-adhesion responsive to temperature. Further, two kinds of algae with different shape/size were incubated to investigate the adhesion behavior on bare and PSPMA-modified fibers arrays. The experiment results show that both the surface topography and chemical composition can influence algae attachment, while the adhesion amount is highly related to the effective attachment points of the cells on the surface. The surface physicochemical properties would weak the cell adhesion more significantly than that of surface structure but which is also not be ignored.

2. Result and Discussion

2.1. Morphology of the PP Nanofibers Arrays

As shown in Figure 1a, the fibrillar PP arrays were fabricated by a hot pressing molding technology using 400 nm polycarbonate (PC) membrane filter as template (Figure S1, Supporting Information). A series of nanofibers arrays with different structural distribution was prepared, with reasonable adjusting the molding time. The morphology of the prepared arrays was characterized by field-emission scanning electron microscope (FESEM). As shown in Figure 1b, all of the prepared arrays possessed well-defined fibers with diameter ≈ 450 nm and presented with ordered topographies. Array A with fibers density of $\approx 0.57 \times 10^6 \text{ mm}^{-2}$ but an aggregation conformation on top, array B with fibers density of $\approx 1.06 \times 10^6 \text{ mm}^{-2}$ and hollow

structure on top, array C with fibers density of $\approx 0.67 \times 10^6 \text{ mm}^{-2}$ and filled structure on top and array D with low fibers density of $\approx 0.26 \times 10^6 \text{ mm}^{-2}$ and aggregation structure on top. As shown in array A, fibers bonded with each other form big valley-like intervals, while freestanding forms small and uniform fibers intervals for array B. By contrast with array B, the freestanding fibers intervals increased for array C. For array D, the fibers bonded with each other form pyramidal-like fibers bundles, where left ordered inverted-pyramid intervals.

2.2. Surface Wettability and Water Adhesion in Air

The as prepared PP fibers of array A, array B, and array C were all with concave shape top except array D. The static contact angle was $\approx 98^\circ$ of flat PP substrate, $\approx 130^\circ$ of array A, $\approx 145^\circ$ for array B, $\approx 155^\circ$ for array C, and $\approx 161^\circ$ for array D (Figure S2, Supporting Information). As shown in Figure 2a, both array A and array B showed strong adhesion to water, water droplet could easily hang on the wall without dropping for 90° or even 180° flip. Array C presented with weak adhesion with high sliding angle of $\approx 17^\circ$, while array D kept superhydrophobic and self-cleaning with low sliding angle of $\approx 7^\circ$. 10 μL water droplet was chosen as the probe to detect the adhesion force on these structured fibers array surfaces. The adhesive forces were measured by a high sensitivity microelectromechanical balance system, which accurately recorded the water droplet approaching, contacting, and leaving the array surface. As shown in Figure 2b, the force–distance curves, recorded before and after the water drop (10 μL) contacted the surfaces, five different points were chosen for testing on each substrate (including the flat). As shown in Figure 2c, the flat substrate showed a high water adhesion force of $\approx 450 \mu\text{N}$ and then reduced gradually from array A, array B, to array

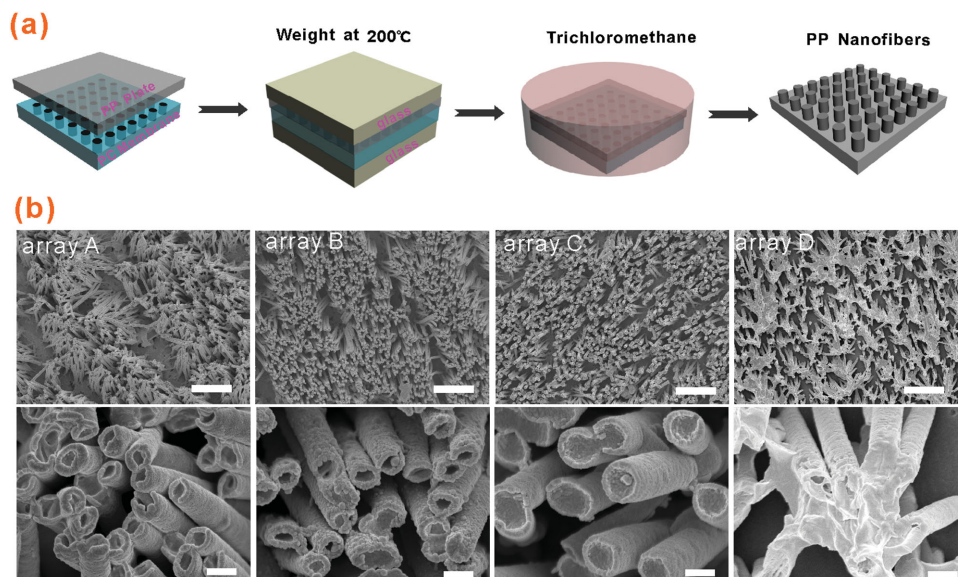


Figure 1. (a) Schematic illustration of the fabrication process of polypropylene fibrillar arrays. (b) Field-emission scanning electron microscope (FESEM) images of the four different kinds prepared polypropylene (PP) fibers arrays with different surface geometry morphology, array A, array B, array C, and array D. (The scale bars from top to down are 10 μm , 500 nm).

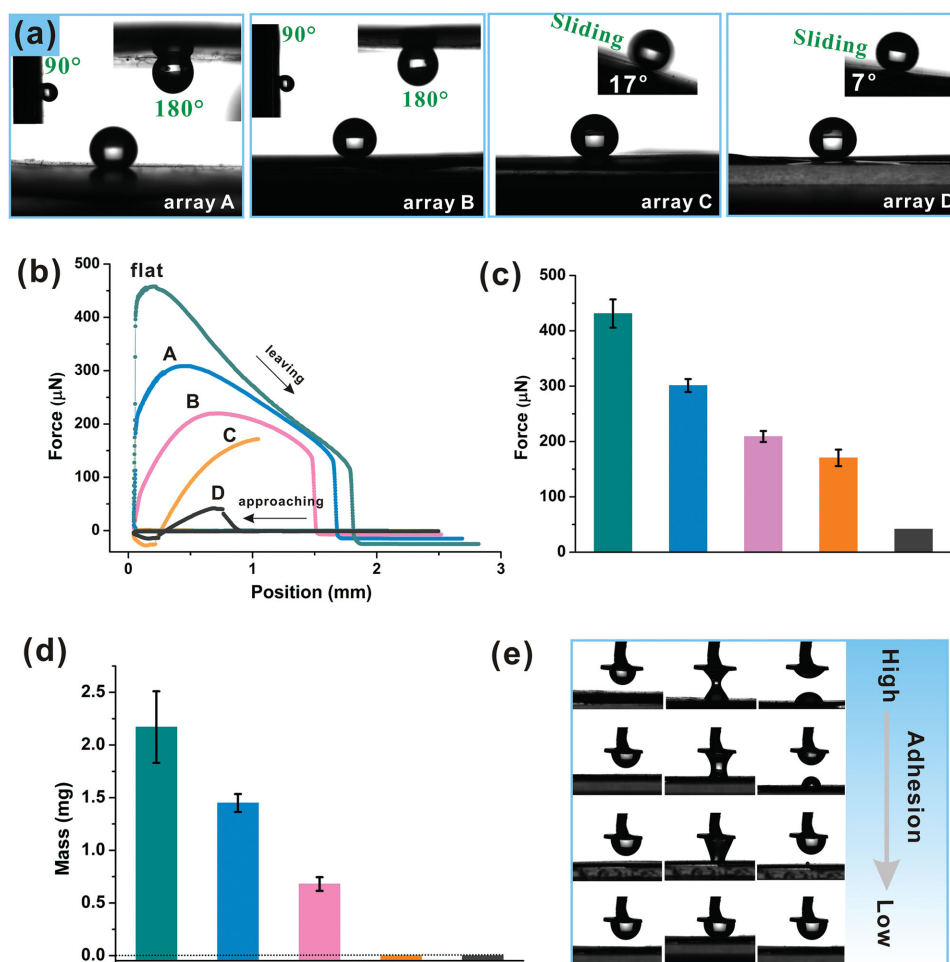


Figure 2. Adhesion measurement of the PP fibers array with different fiber orientations. (a) Photos for 5 μL water droplet during different adhesion states, as hang on while tuned with 90° or even 180°, or sliding away. (b) Force–distance curves of the water droplet (10 μL) approaching, contacting, leaving, and separating process with substrates. (c) Histogram of corresponding adhesive amount valued from the curves. (d) Residual mass of the water droplet after measurement. (e) Photographs taking to stimulate the corresponding stages during the measuring process by a CCD camera (From top to bottom: flat substrate, array A, array B, array D).

C, while array D behaved the low adhesion force of ≈ 40 μN for water. After testing, no residual water was found both for array C and array D (Figure 2d). By contrast, the residual mass of adhered water on flat PP substrate could reach ≈ 2.1 mg, while reduced to 1.4 mg on array A and 0.7 mg on array B (Figure 2d). Correspondingly, these dynamical adhesion processes could be traced and stimulated in sight of an optical camera (Figure 2e).

2.3. Wetting Models

The adhesion evolution on four kinds of arrays could be attributed to different wetting models. As shown in Figure 3, four different kinds of wetting models were suggested to match array A, array B, array C, and array D. Among, array A, array B, and array C possessed well-fined fibers with concave shape top, which benefits strong capillary force,^[22] while this special structure disappeared on array D. Both the top capillary force

and intervals roughness influence the wetting state. In fact, PP is well known as a low surface energy material, and existing of the micro/nanotopography would effectively repel the adhesion of water. The as-prepared arrays A showed wide valley-like fibers intervals of ≈ 10 μm with capillary waters, leading to high adhesion of water with stable Wenzel's state. Array B with freestanding fibers topography showed uniform fibers intervals of ≈ 1 μm , so the surface was in "transition state" with hydrophobic but adhesive attributing to the reduced fraction of air among intervals. This array could be used to transfer water droplet from superhydrophobic surface to superhydrophilic surface (Figure S3, Supporting Information). With the enlargement of the intervals size to ≈ 3 μm , top capillary force decreased gradually, and array C was in state "gecko state" with superhydrophobic but weak adhesion. For array D, the capillary force from fibers' concave shape top disappeared, and air fraction among the conical fibers bundle decreased to the lowest, the surface was in stable "Cassie's state" with superhydrophobic and self-cleaning.

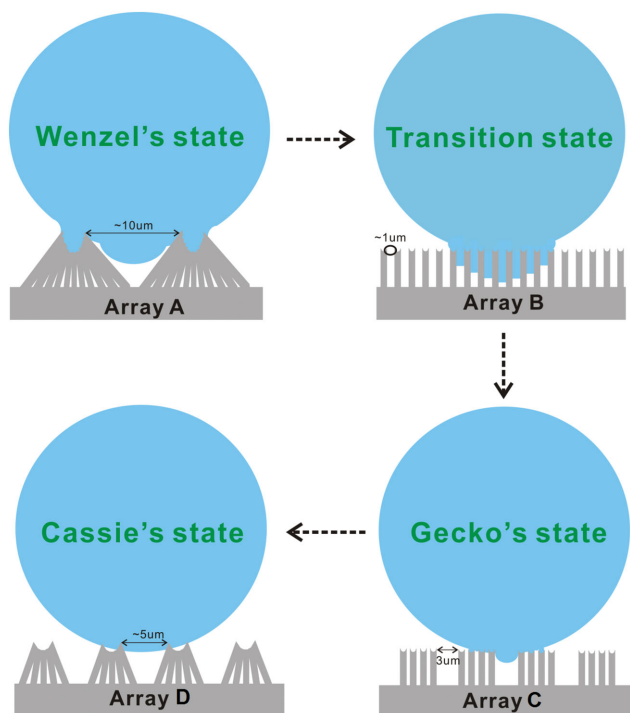


Figure 3. Possible wetting model for four kinds of PP arrays with different structure topographies.

2.4. Wetting Evolution from Air to Underwater

As depicted above, the as prepared PP fibers array D would present highly self-cleaning property to water droplet with sliding angle of less than 10° . Further, it was found that array D surface showed a considerable increase in its repellency to a series of water-soluble liquid droplets. As illustrated in Figure 4a, the juice, coffee, milk, and blood droplets (10 μ L) kept round ball on array D with corresponding contact angle (CA) of 152.68° , 152.61° , 150.31° , 147.12° . This means that the array D could almost show a lotus effect (CA $> 150^\circ$). As shown in Figure 4b, the juice (10 μ L) droplet has low sliding angle of 9° with well self-cleaning. While the coffee, milk, and blood (10 μ L) droplets could easily slide away with sliding angle of 12.6° , 17.8° , 30.5° , respectively. These results showed that the as prepared superhydrophobic PP fibrillar surface could almost demonstrate the self-cleaning behavior as other artificial materials.^[15,23,24]

However, different from the water-soluble contaminations, the as-prepared array D displayed lipophilicity for dichloroethane (DCE), organic silicone oil (OSO) both in air and underwater (Figure 4c). This evoked our attention for thinking of those seabirds in gulfs, whose superhydrophobic feathers were often contaminated by the leaked oil in water. Once the feathers were coated oil, it is difficult to remove them away and life was threatened. So could these self-cleaning materials in air also display effect underwater (in fresh water or seawater)? The answer is worth thinking about carefully. Besides, the real environment underwater (especially in ocean) is extremely complex, where many living organisms are habit while form bad aggression to marine equipments.

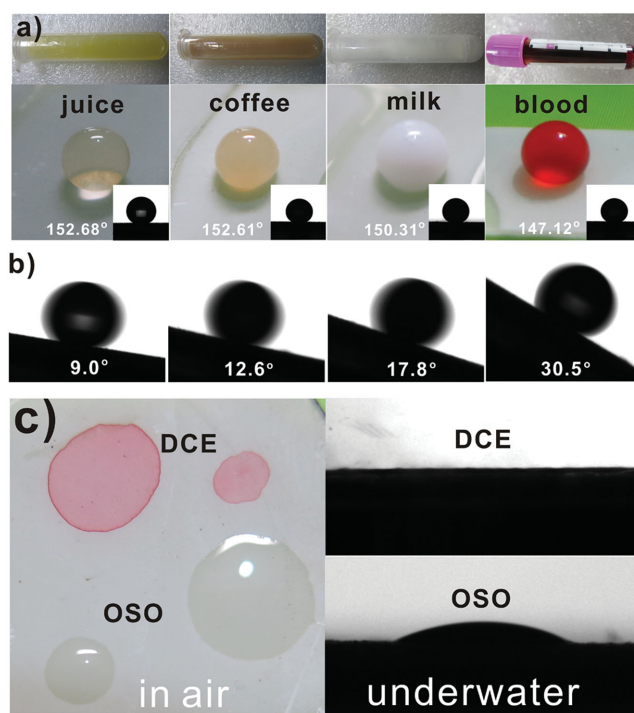


Figure 4. Wetting of water-soluble droplets and oil droplets on PP array D. a) Photos of juice, coffee, milk, and blood droplets (10 μ L) on array D. b) Photos of corresponding sliding angle of juice, coffee, milk, and blood droplets (10 μ L) on array D. c) Photos of dichloroethane (red-DCE), organic silicone oil (OSO) droplets on array D in air and underwater.

One possible strategy is learning from fish whose skin could achieve underwater oleophobicity and keep clean against marine organisms, due to existence of high surface energy hydrophilic chemistry composition and micro/nanostructure on skin surface.^[18]

2.5. Preparation of Hydrophilic PP Fibrillar Array by SI-ATRP Using Mussel-Inspired Initiator

An effective approach to repel the oil adhesion on these structured fibers arrays was changing the surface wettability by decorating hydrophilic polymer. Array C was chosen for surface modification and testing for oil droplet and cell adhesion due to ordered fibers distribution. Inspired by the mussel that can tightly attach to the ships, catecholic groups have been developed to modify the surfaces of various inorganic and organic materials.^[25–30] Here, the PP fibers were first modified with dopamine initiator,^[21] and then grafted functionalized polymer brushes via the SI-ATRP.^[31] As shown Figure 5a, successful initiator immobilization was proved by the emergence of N 1s signal and Br 3d signal from X-ray photoelectron spectroscopy (Figure S4, Supporting Information). Then poly (3-sulfopropylmethacrylate potassium salt) (PSPMA) was grafted onto the fibers by SI-ATRP, which were ascertained by emergence of S 2s and S 2p signal (Figure S5, Supporting Information). Further, successful PNIPAAm (N-isopropyl acrylamide) grafting on PP fibers was indicated by the emergence of the N

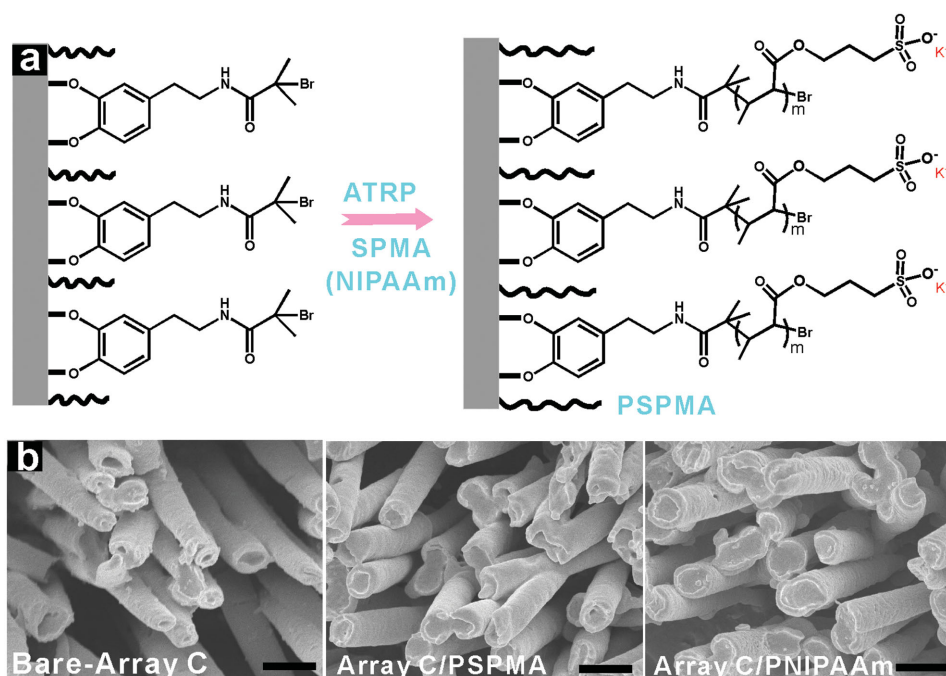


Figure 5. Schematic diagram for modification of the fibrillar PP arrays and corresponding FESEM images. a) Schematic diagram for the assembly of mussel-inspired initiator (Dop-Br), then followed by grafting hydrophilic polymer brushes (PSPMA) and temperature-responsive polymer brushes (PNIPAAm) by surface-initiated atom transfer radical polymerization (SI-ATRP). b) Corresponding FESEM images of bare PP fibers arrays (left), after grafting PSPMA (middle) and PNIPAAm (right). (The scale bar is 1 μm).

1s signal (Figure S6, Supporting Information). SIP on flat PP substrates were performed in a similar way. FESEM images showed that the polymer brushes grafting fibers become smooth with lightly increasing of the size comparing to the unmodified (Figure 5b). The modification only changed the chemical composition of interface with little effect on surface roughness.

2.6. Low Oil Adhesion Underwater on Polyelectrolyte Brushes (PSPMA) Decorated Structured PP Arrays

A 20 μL oil droplet of 1,2-dichloroethane with surface tension of 24.15 mN m^{-1} was chosen as probe to characterize interfacial contact angle and adhesion force underwater. After surface-induced grafting of PSPMA, the flat substrate showed oleophobicity with CA of 121.70° (Figure 6a), while composite array showed super oleophobicity with CA of 155.68° , which attributed to the existence of the structured roughness in the solid/oil/liquid interface and grafting hydrophilic polymer brushes (Figure 6b).^[32] The adhesive forces were measured by same microelectromechanical balance system with 10 μL 1,2-dichloroethane attached on metal ring, which directly allowed the oil droplet to contact the surface and then leave. The force–distance curves were recorded from contacting to leaving of oil droplet. As shown in Figure 6c, the oil droplet showed high adhesion on PSPMA-modified flat substrate, while showed low adhesion on composite array surface (Figure 6d). The inset photographs of oil droplets were taken to stimulate the corresponding stages during the measuring process. PSPMA was

well known as one of highly hydrated polyelectrolyte brushes, has been widely investigated for biolubrication,^[33–36] and anti-fouling.^[37–43] Hydrated polyelectrolyte brushes would form dense hydration layer,^[44] which resisted the approach of the oil droplet, both on array and flat substrate. However the surface topography and chemical composition would both influence the wetting of solid/oil/liquid. Considering the gravity, the oil droplet would be more sufficiently contacting flat surface though existed hydration layer (Figure 6e). By contrast, hydrated fibers would effectively repel the oil to contact the surface due to existing interval space (Figure 6f). This is consistent with the accepted result that surface roughness contributes significantly to enhancing liquid repellence.^[45,46]

The existing array structure will highly enhance self-cleaning capability, which could be clearly recorded with compression of large volume oil droplet using a camera. As shown in Figure 7a, 50 μL oil droplet was placed between two pieces of plates for polyelectrolyte brushes (PSPMA) decorated arrays. It was found that when two plates were pressed together with 1 mm gap, the oil droplet could slip away easily from the interface under external compressive forces of 118 Pa. The grafted polyelectrolyte brushes provided effective hydration layer as boundary lubrication film, which facilitates the oil droplet slipping from the interface. Figure 7b clearly showed the pressing, shaping, and releasing process of the oil droplet under asymmetrical shear. In both cases, no residual oil droplet was observed, and the interface maintained original cleaning state. In general, increasing the volume of the oil droplet would facilitate the adhesion due to the rising gravity and contact area.^[47] Interestingly, the as-prepared arrays kept strong repelling to

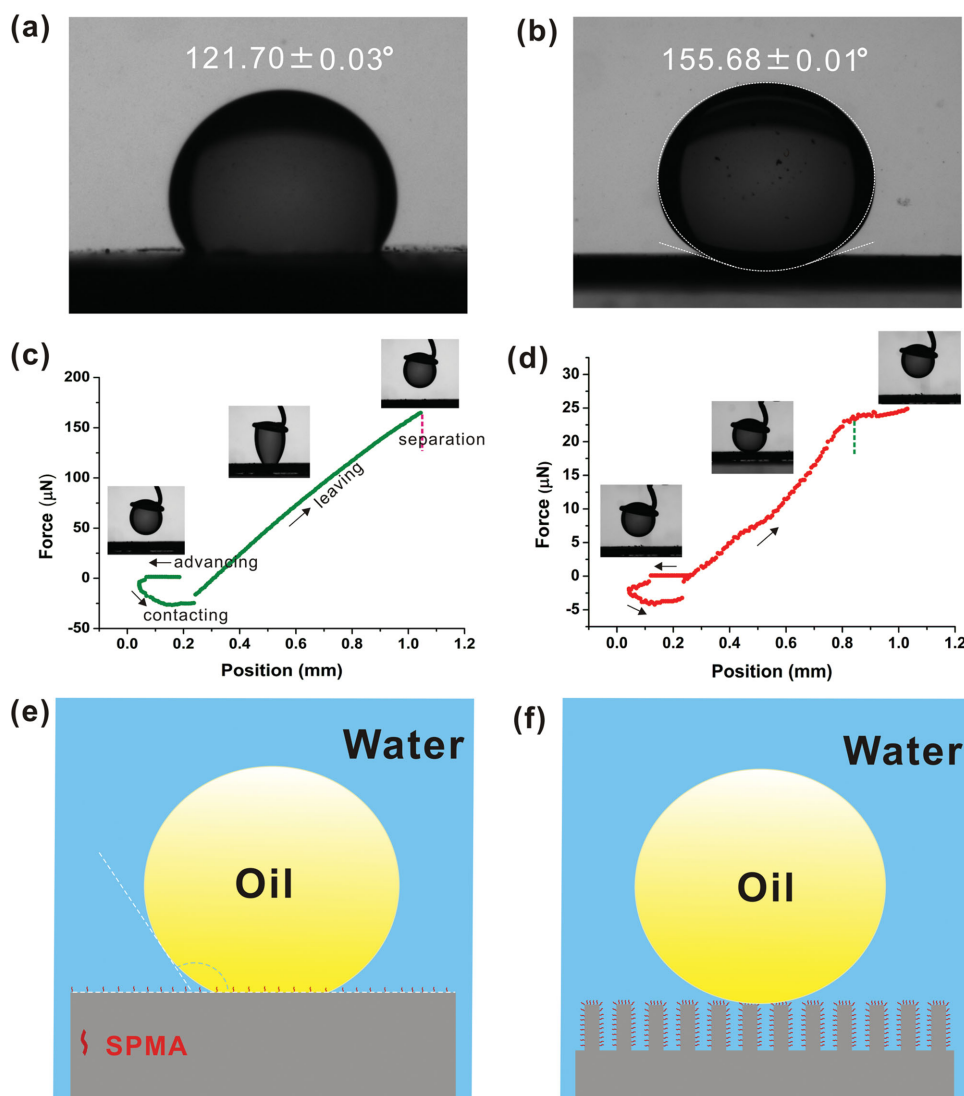


Figure 6. Contact angles of 20 μL oil droplet of 1,2-dichloroethane on (a) PSPAM-modified flat surface, and (b) PSPAM-modified array surface. Force–distance curves before and after PSPAM-modified flat surface contacted with the oil droplet (10 μL) on (c) PSPAM-modified flat surface, and (d) PSPAM-modified array surface. The inset photographs of the oil droplets were taken to stimulate the corresponding stages during the measuring process. Possible contact mechanism for oil droplets on (e) PSPAM-modified flat surface, and (f) PSPAM-modified array surface.

50 μL oil droplet, which means that the composite interface could be presented with highly self-healing as marine plants and animals.

2.7. Switchable Underwater Oil Droplet Adhesion

The PNIPAAm-coated array C exhibited reversible switching from superhydrophobic state to superhydrophilic state (Figure S7, Supporting Information). Subsequently, controllable wetting switching was performed on poly(*N*-isopropyl acrylamide) (NIPAAm) decorated PP fibers array. For PNIPAAm, the conformation of its molecular chains is highly stretching and forms intermolecular hydrogen bonds with water molecules below its LCST (lower critical solution temperature) of 32 $^{\circ}\text{C}$, while collapse and forms intramolecular hydrogen

bonds through amido bond above its LCST of 32 $^{\circ}\text{C}$.^[48] In this case, the composite array showed hydrophilic in air below LCST and kept oleophobic underwater, while hydrophobic above LCST in air and presents adhesion to oil droplet underwater. With the temperature increasing, WCA increased from $49.86 \pm 2.24^{\circ}$ to $64.01 \pm 2.04^{\circ}$ on PP-NIPAAm in air (Figure S7a, Supporting Information), while the OCA decreased from $131.49 \pm 0.50^{\circ}$ to $125.84 \pm 0.22^{\circ}$ underwater (Figure S7b, Supporting Information). For PPNF–NIPAAm fibers array, with the temperature increasing, WCA converted obviously from $17.97 \pm 0.18^{\circ}$ to $112.15 \pm 0.08^{\circ}$ in air (Figure S7c, Supporting Information). Figure S6d (Supporting Information) showed that the composite array changed from superoleophobicity to oleophobicity, with the OCA decreasing from $153.80 \pm 0.1^{\circ}$ to $134.53 \pm 0.16^{\circ}$.

Then we detected the adhesion force of oil droplet on PP–NIPAAm substrate and PPNF–NIPAAm array by using 10 μL

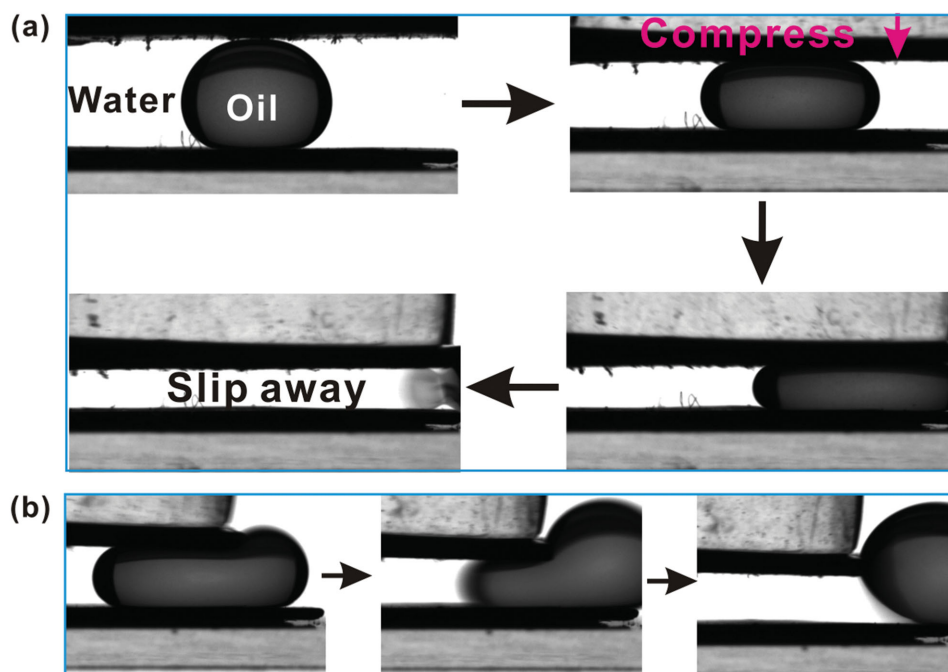


Figure 7. 50 μL oil droplet movement under external forces on PSPMA-grafted PP fibrillar surface. (a) Show the oil droplet in an untreated state, forced state, and sliding state, and slipping state, respectively. (b) Shows in situ the oil droplet shaping and escaping process under vertical shear force.

1,2-dichloroethane as probe. As shown in **Figure 8a**, oil droplet could well hang on composite array vertically overturn of 180° above the LCST of NIPAAm, while slid away easily in a low tilt angle of 4° below the LCST (**Figure 8b**). **Figure 8c** showed the in situ force–distance curve of the oil droplet above the LCST, which presented with high adhesion to composite array. In contrast, the adhesion decreased obviously below the LCST (**Figure 8d**). Above the LCST of PNIPAAm, collapse of the polymer chains would lead to a dehydrated state around the fibers. As a result, the composite array surface was in a typical Wenzel's state with high oil adhesion (**Figure 8e**). Below the LCST, the polymer chains around the fibers were in a hydrated stretched state. As a result, the composite array surface was in a typical Cassie's state with low oil adhesion (**Figure 8f**).

3. Cell Adhesion

As mentioned above, these structured PP arrays presented gradient wetting evolution from high adhesion to self-cleaning, while exhibited excellent superoleophobicity and controllable oil-droplet adhesion property underwater after grafting functional polymer brushes. However, the accepted fact is that many creatures in ocean exposing to complex habitat environment where is rich of many living fouling organisms, different from oil droplet. Interestingly, these creatures kept highly clean under such complex environment. By contrast, boats and ships without life could easily be aggressed and fouled. Many research results showed that both surface chemical composition and microtopography on these creatures, such as shark skin,^[49] *N. septentrionalis* skin,^[18] mollusk shells,^[50] or coral,^[51] were responsible for antifouling, but the underlying mechanism is

still controversial and unclear. So it is still highly significant to systematically investigate the possible mechanism of surface composition and structure on preventing fouling.

Here, two different kinds of marine algae cells with different shape and size were incubated for fouling test considering the matched fibers scale of PP arrays. One kind is *Chlorella* sp. with ball shape, showed similar diameter distribution of $\approx 3 \mu\text{m}$. The other kind is *Navicula* sp. with asymmetrical structure distribution of ship, kept $\approx 4 \times 7 \mu\text{m}$ size. In our experiment, algae adhesion on bare PP arrays and as-modified PP arrays were tested under artificial seawater at room temperature. All samples were individually placed in culture dish and filled with 10 mL of algae culture suspension (the *Chlorella* sp. was adjusted about $8.15 \times 10^6 \text{ cell mL}^{-1}$, while *Navicula* sp. density was $1 \times 10^6 \text{ cell mL}^{-1}$) and then grown for 24 h. Ten images of each of the three replicate samples were photographed to quantify the number of attached algae. It is worth exploring if there is some certain relationship between the organisms fouling and the interfacial wetting, well-distributed structures and physicochemical properties on these arrays in static incubation, without considering the effect of fluid dynamics on cell settlement.^[52]

3.1. *Chlorella* sp. Adhesion

The structured fibers array with low surface energy kept hydrophobic state. As shown in **Figure 9a**, *Chlorella* sp. attachment was the lowest on the array B, while the most in array A. By contrast, the flat PP substrate presented relative high cell attachment. For array A, the fibers intervals were about $10 \mu\text{m}$, which provide secure valleys for cell settlement. For array B,

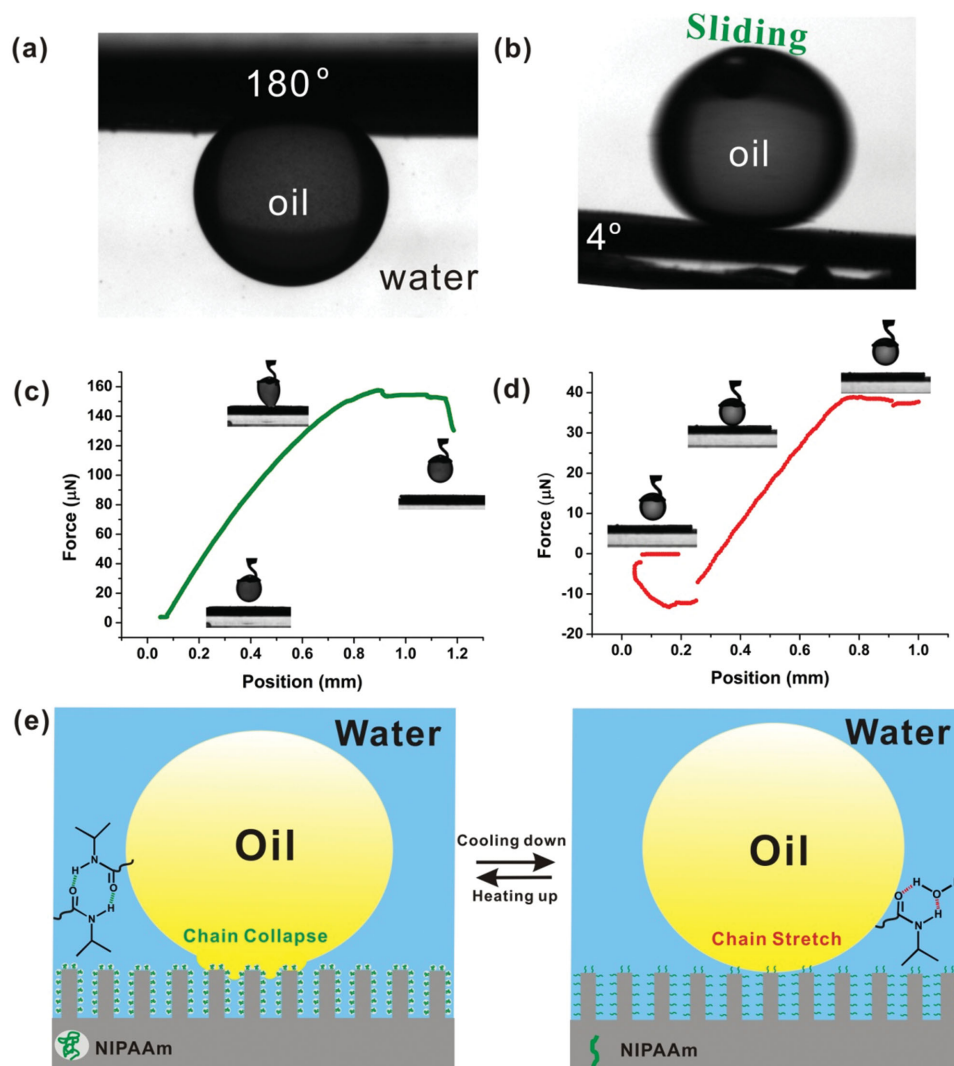


Figure 8. (a) Underwater oil droplet (20 μL) contact angle (OCA) on PNIPAAm grafting array surface when turning with 180° in vertical direction at 45°C ($>\text{LCST}$). (b) Underwater oil contact angle (OCA) on PNIPAAm grafting array surface when slipping away with low angle of 4.0° at 25°C ($<\text{LCST}$). Force–distance curves of the oil droplet (10 μL) approached and left the PNIPAAm grafting array surface (c) at 45°C ($>\text{LCST}$), (d) at 25°C ($<\text{LCST}$). The inset photographs of the oil droplets were taken to stimulate the corresponding stages during the measuring process. (e) Possible wetting model of oil droplet (left) above and (right) below the LCST of PNIPAAm.

the effective fibers intervals is $\approx 1\ \mu\text{m}$, which is about threefold size of the *Chlorella* and provided with possibility of the most effective attachment points. For array C, the fibers interval is almost matched with the *Chlorella* and the cell tended to settle in the intervals except on the fibers surface. For array D, the increased fibers intervals ($\approx 5\ \mu\text{m}$) led to locking of much more cells to valleys. This observed result is highly coordinated with the contact-points theory.^[53] However, no obvious relativity was observed between the *Navicula* sp. attachment and surface wetting, which is different from the relative reports that surface wettability from different topographies may be relative to cell settlement and attachment.^[54,55] For example, array D presented with stable Cassie's state and was found with low sliding angle of 7° , while the cell attachment was not lowest. Because the physicochemical properties of the surface was impacted after immersing in the cell incubation solution,

which would change the wetting at some extent.^[56] Besides, different from wetting state of small volume liquid droplet in air, once the substrates immersed, the existing liquid pressure would prompt the direct contact of liquid with surface, so the swim cell could also easily attach to the surface. After grafting with negative charged polyelectrolyte brushes (PSPMA), all the surfaces presented very low cell adhesion. The modified plat substrate presented with relative high cell attachment comparing with other well-constructed fibers array substrates, the settlement density of *Chlorella* sp. was the lowest on the modified array B.

As reported, the substrate's surface energy and roughness would both impact the cell fouling.^[57] For surface energy, no certain adhesion generalities were found due to the variety of studied organisms, experiment conditions, and different physicochemical properties of cell, while the roughness is

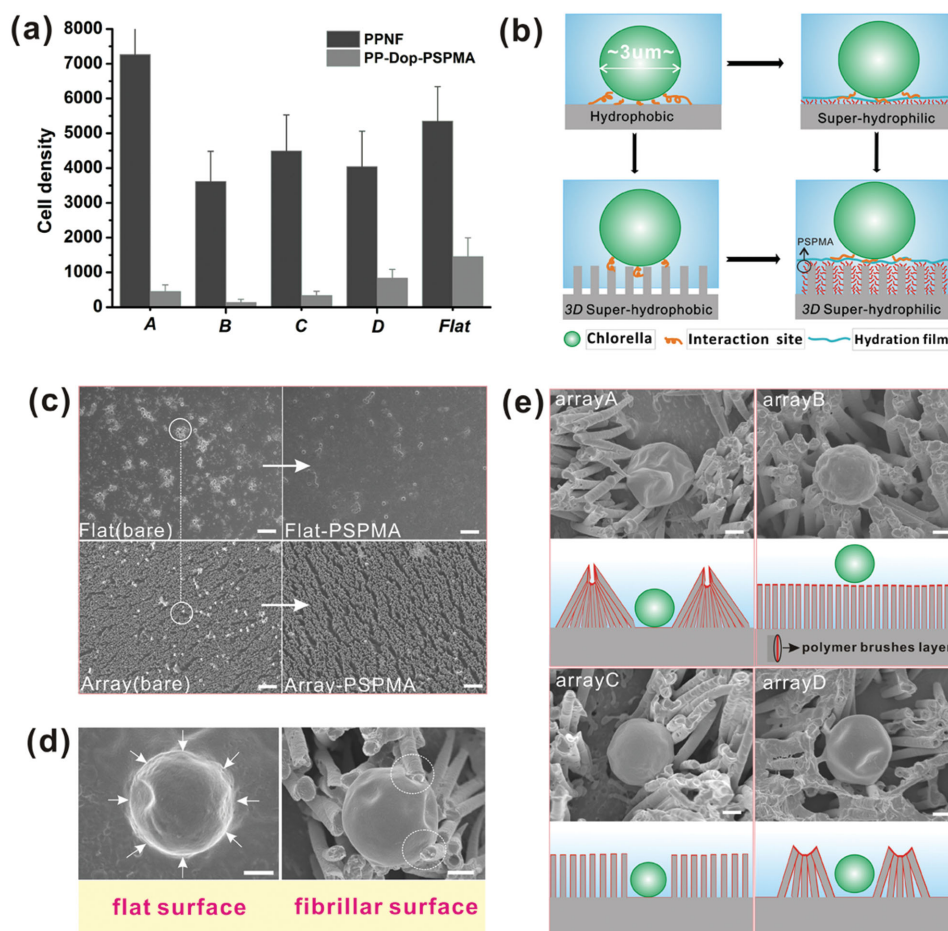


Figure 9. Adhesion testing of *Chlorella* sp. and correspondingly attachment models. (a) Histogram of statistically results of cell adhesion amount. (b) Possible attachment mechanism of *Chlorella* on bare PP flat substrate, PSPMA-modified flat substrate, bare PP fibers array, PSPMA-modified PP fibers array, and (c) corresponding FESEM images. (d) High magnified FESEM images of clear adhesion state for *Chlorella* on bare flat PP substrate and structured PP fibers array. (e) High magnified FESEM images for incubation *Chlorella* cell in PSPMA decorated structured fibers intervals and corresponding settlement models. (The scale bars in Figure (c) is 20 μm , Figure (d), (e) is 1 μm).

highly related to the effective number of attachment points of cell.^[58–60] As shown in Figure 9b, the *Chlorella* sp. approaches to surface and may attach on effective points by secreted extracellular polymeric substances (polysaccharides and proteins).^[61] The low surface energy of PP means that strong hydrophobic interaction would take place between the substrate and nonspecifically distributed hydrophobic pocket in polysaccharides and proteins regions of the cell.^[62,63] So both flat and structured surfaces presented with high *Chlorella* attachment, and the attachment amount related to the effective contact points on each array. After modification with PSPMA, the surface energy of the substrate increased, with high interfacial negative charges density and effective hydration layer,^[64] which will prevent the approach of negative charged *Chlorella* due to double electrical layer repulsion effect from polar interaction.^[65] As a result, the cell attachment was weakened effectively both for flat and array surfaces (Figure 9c). By contrast with array substrate (Figure 9d, right), the flat substrate would create more contact points with large contact area for cell attachment due to the lacking of effective supporting site (Figure 9d, left). As shown in Figure 9e,

most of the *Chlorella* sp. would be unnecessarily locked into the valleys of PSPMA-array A attributing to effective hydration layer after modification. By contrast, the *Chlorella* sp. could only settled on fibers top of the modified array B with most favorable contact points, endowing the weakest attachment. However, the *Chlorella* sp. could settle on either fibers top or fibers interval of PSPMA-array C, resulting in increasing attachment amount. For PSPMA-modified array D, almost all of the *Chlorella* approached to the surface and then settled on the inverted pyramid fibers clusters intervals, so the number of cell adhesion was further increased.

3.2. *Navicula* sp. Adhesion

As shown in Figure 10a, the *Navicula* sp. presented with high adhesion on bare arrays and flat substrate, while presented with low attachment amount after grafting polyelectrolyte brushes (PSPMA). Even though *Navicula* sp.'s size is larger than the interval scale of bare array B, the statistical results did not show that reduced tendency for cell adhesion. The flat substrate

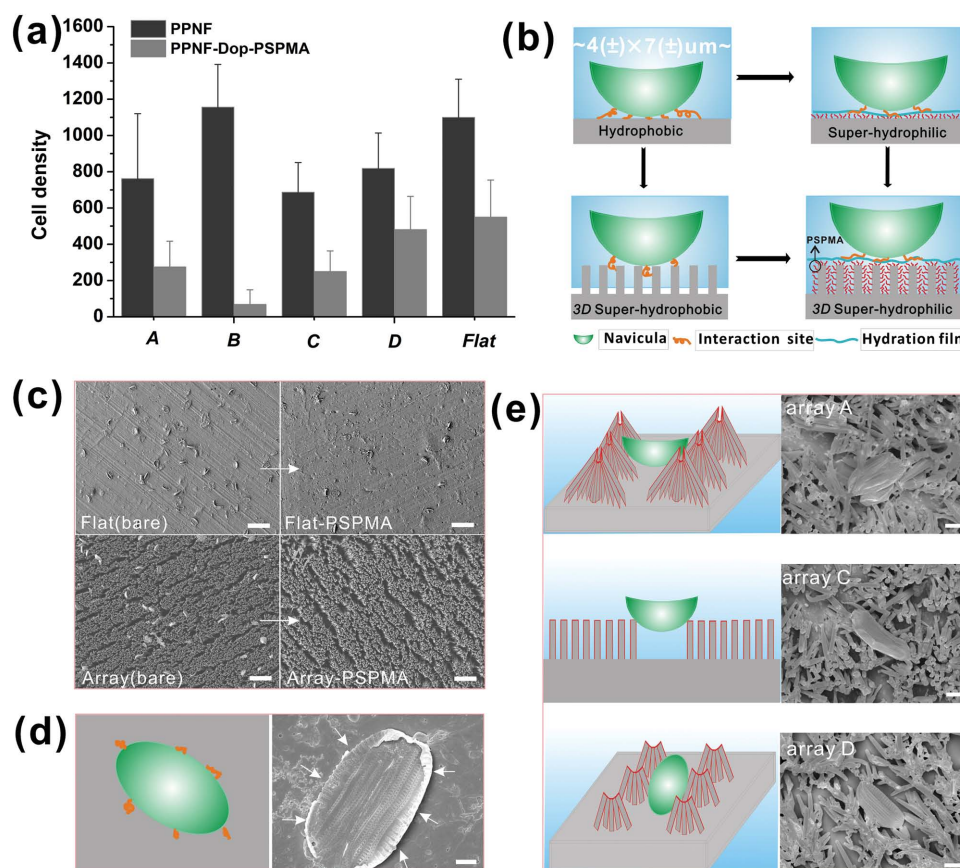


Figure 10. Adhesion testing of *Navicula* sp. and correspondingly attachment models. (a) Histogram of statistically results of cell adhesion amount. (b) Possible attachment mechanism of *Navicula* sp. on bare PP flat substrate, PSPMA-modified flat substrate, bare PP fibers array, and (c) corresponding FESEM images. (d) High magnified FESEM images of clear adhesion state for *Navicula* sp. on bare flat PP substrate. (e) High magnified FESEM images for incubation *Navicula* sp. cell in PSPMA decorated structured fibers intervals and corresponding settlement models. (The scale bars in Figure (c) is 20 μ m, Figure (d),(e) is 1 μ m).

exhibited decreasing adhesion amount comparing with unmodified array B, while array A, array C, and array D presented with the relatively low adhesion comparing with flat substrate. One possible reason is that *Navicula* sp. presented with irregularity of size distribution, which means the adhesion could happen along each direction of the cell. For example, they could contact fibers top with the ship's bottom along the short-axis direction or embed into the fibers intervals directly along the long-axis direction. At this point, high density fibers array provides much more contact points for *Navicula* sp. Also, no correlation between *Navicula* sp. adhesion and interfacial wetting on unmodified arrays was found. The whole trend is a statistically significant reduction in spore attachment on PSPMA grafting PP surfaces. Interestingly, it presented with same adhesion tendency as *Chlorella* sp. The modified array B presented with lowest adhesion, while the flat substrate kept high adhesion. The same adhesion mechanism of *Chlorella* could also be used for analyzing the attachment of *Navicula* sp. (Figure 10b). One of the most obvious differences of attachment mechanisms is that *Navicula* sp. has much more contact styles than *Chlorella* sp. both for unmodified and modified surfaces, due to its irregularity of the cell shape distribution. As shown in Figure 10c, the cell adhesion amount reduced obviously both for flat and array substrate after grafting

polyelectrolyte brushes (PSPMA). Due to lacking of effective supporting sites, much bigger contact area would facilitate *Navicula* sp. to settle on the flat surface (Figure 10d). After modification, the *Navicula* sp. could also settle into fibers intervals of the array. As shown in Figure 10e, the *Navicula* sp. could easily settle into the valleys of array A along each direction, while partial settled into the intervals of array C and completely fell into the bottom of the intervals of array D along the long-axis direction.

According to our statistically results, two kinds of cells more tended to settle on these low surface energy but structured PP fibers array surface, which could be speculated as strong hydrophobic interaction. By contrast, these arrays showed low attachment after grafting polyelectrolyte brushes (PSPMA), which enable to form high interfacial negative charges density and effective hydration layer to prevent the approach cell from polar interaction.^[65] The adhesion amount of *Chlorella* sp. and *Navicula* sp. on unmodified and modified PP arrays was highly depended on the effective attachment points at the interface. When the cell size was larger than the intervals scale of the fibers, the topography exhibited low adhesion amount due to little attachment points. When the fibers interval was larger than the cell, abundant adhesion would take place due to sufficient attachment points. However this attachment regularity present

slight difference depended on cell type. Comparing with PSPMA grafting flat substrates, all PSPMA-modified arrays surfaces exhibit excellent anti-adhesion for *Chlorella* sp. and *Navicula* sp., which is due to the existed structure effect. So both the surface well-defined topography and chemical composition would influence the attachment for two different kinds of cells. The surface physicochemical properties would weak the cell adhesion more significantly than surface structure but which is also not be ignored.

4. Conclusion

Inspired by the nature, four kinds of PP nanofibers arrays with different surface geometries were fabricated using a universal molding method, the wettability and adhesion behavior of these arrays were evaluated both in air and underwater. The adhesion force to water droplet reduced gradually from flat substrate to arrays A–D, due to varied fibers interval and special top shape. Among them, array C shows superhydrophobic but with high adhesion, while array D kept highly repellent and low adhesion to a series of liquid including water, juice, coffee, milk, and blood, indicating excellent antifouling performance of the surface in air. Due to obvious lipophilicity both in air and water, they may not well suitable for the underwater antifouling, even though these fibers arrays showed well superhydrophobicity in air. Inspired by fish, we have gained functional fibrillar PP array by grafting hydrophilic polyelectrolyte brushes (PSPMA) via SI-ATRP, and the as-prepared composite array showed stable superoleophobicity underwater. Subsequently, the antibiofouling properties of PSPMA grafting four PP arrays were evaluated using two kinds of algae cells (*Chlorella* sp. and *Navicula* sp.) with different shape and size. The experiment results showed that these PP-PSPMA arrays can mitigate biofouling effectively, but are influenced by the surface topography and chemical composition, and the anti-adhesion performance is highly related to attachment points of the cells on the surface. When the cell size was larger than the intervals scale of the fibers, the substrates exhibited low adhesion amount due to little attachment points. When the fibers interval was larger than the cell, large amount of algae adhesion took place due to sufficient attachment points. This investigation of the underlying correlation between the fouling mechanism and the interfacial wetting, topography, physicochemical properties will be constructive to develop efficient and low-cost self-cleaning marine antifouling materials.

5. Experimental Section

Preparation of the Well-Structured PP Nanofibers Arrays: The fibrillar arrays were fabricated by casting a 500- μm -thick PP film in a vacuum oven at 200 °C into a 20- μm -thick PC track-etched membrane filter (Whatman) containing 400-nm-radius pores, as shown in Figure 1a. Samples were molded in a constant hydraulic press maintained by 1 kg load at 200 °C in $\approx 10^{-5}$ Pa vacuum atmosphere at power metallurgy instrument with different heating time, and then followed to cool with 45 min before being released. The PC membrane was etched in trichloromethane to release the molded fibers, and then rinsed with isopropyl alcohol and air dried.

Initiator Anchor and SI-ATRP: The as prepared PP substrates were rinsed by repeated washing with absolute ethanol, activated by O_2

plasma for 5 s. Then the substrates were immersed in 1.5 mg mL^{-1} solution (ultrapure water: ethanol = 4:1, v/v) of initiator for 24 h in the dark. Then polymer brushes were grafted from the initiator anchored substrates by SI-ATRP. SPMA ((3-sulfopropylmethacrylate potassium salt)) (3 g, 55.23 mmol), Bipy (2,2-bipyridine) (0.039 g, 3.35 mmol), and CuBr (0.0175 g, 1.12 mmol) were dissolved in a mixture of methanol/water (6 mL 1:2 (v/v)) to prepare the reaction solution. NIPAM (N-isopropyl acrylamide) (3.78 g, 33 mmol), PMDETA (N,N,N',N',N''-pentamethyldiethylenetriamine) (210 μL), and CuBr (Copper(I) bromide) (0.048 g, 0.33 mmol) were dissolved in methanol/water (9 mL, 1/1, v/v). The polymerization reaction was carried out at room temperature under a nitrogen atmosphere for 1 h. After polymerization, the samples were rinsed with pure water and dried in a nitrogen jet.

Cell Cultivation and Adhesion Testing: *Chlorella* sp. and *Navicula* sp. were obtained from Institute of Hydrobiology and cultured in Erdschreiber medium and Dunaliella medium in the box with constant temperature and humidity for attachment experiments. The cell density of *Chlorella* sp. is approximately 8.15×10^6 cells per mL, while the *Navicula* sp. is approximately 1×10^6 cells per mL. The samples were immersed into cell incubation solution for 24 h, and then were rinsed by dipping each treatment in a new beaker of artificial seawater three times to remove unattached cells. Cells were counted from 30 random fields of view.

Instruments and Characterization: Array morphologies were observed in FESEM (JSM-6701F, Jeol, Japan) at 5 kV. Static contact angles of water were acquired using a DSA-100 optical contact angle meter (Hamburg, Germany). 5 μL of deionized water was dropped on the samples using an automatic dispense controller. The adhesive force was measured by a high-sensitivity microelectromechanical balance system (Dataphysics DCAT11, Germany). In brief, 10 μL water droplets or oil droplet was attached on a hydrophobic metal ring, which was allowed to approach, contact, and leave the surfaces at a constant speed. Once the water droplet or oil droplet contacted the surface, the force gradually increased and reached the maximum before the droplet broke away from the surface, and the peak data were recorded in the curve as the adhesive force. The chemical composition of the samples was obtained by X-ray photoelectron spectroscopy (XPS) using a PHI-5702 multifunctional spectrometer using Al K α radiation, and the binding energies were referenced to the C 1s line from adventitious carbon at 284.8 eV.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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